

Problem 1

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Problem 1 Let f be a function given by: $f(x) = (\tan^{-1}(x))^2$.

(a) State the domain of f . Write your answer in interval notation.

(b) Find the critical points of f . Show your work!

(c) Show that $f''(x) = \frac{2 - 4x \tan^{-1}(x)}{(x^2 + 1)^2}$.

(d) Use the **second derivative test** to determine whether critical point(s) of f correspond to local minimums, local maximums, or whether the test is inconclusive.